

RUI XING

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TECHNICAL SKILLS

Core Domain	Large-Scale Optimization, GenAI Agents, and AWS ML Infrastructure.
Language	Java, Python, C, C++, Typescript, SQL, Bash
Technologies	AWS (Bedrock, ECS, Lambda, Stepfunction, DynamoDB), CDK, Docker

RELEVANT WORK EXPERIENCE

Amazon	Bellevue, WA
<i>Software Development Engineer</i>	2020 – 2026

Large-Scale Order Quantity Optimization Engine 2025

- Architected and engineered an end-to-end scalable system for large-scale Order Quantity Optimization for Amazon's sourcing team, utilizing a novel Column Generation decomposition framework to manage million-variable resource models, improved order quantity optimization efficiency.
- Developed the core high-performance engine in Java, implementing custom data structures and a selective variable pricing mechanism that minimized memory overhead and reduced model convergence runtime by 9.5%.
- Optimized system throughput by implementing multi-threaded execution for candidate evaluation, achieving a 20% reduction in latency and maximizing hardware utilization during high-load scenarios.
- Designed a robust constraint verification framework to ensure algorithmic fidelity; integrated ML-based heuristics (Linear Regression/Decision Trees) to dynamically predict and optimize execution paths.

Complex Workflow LLM Agent/GenAI Observability 2025

- Initiated a Generative AI project to monitor massive-scale distributed discrete-event simulations, benchmarking near-real-time execution velocity against historical baselines to provide a one-click abnormality detection experience to managers.
- Developed an end-to-end MVP, leveraging AWS Bedrock agentic workflows and Lambda-backed APIs to query heterogeneous sources and synthesize evidence-linked analysis narratives, replacing manual drill-down toil.
- Drove the initiative forward by setting technical direction, documenting the reference workflow, and enabling partner team adoption of a production-ready implementation.

Simulation Fidelity & Sensitivity Analysis 2024

- Designed and developed a sensitivity analysis framework to de-risk an upstream dependency deprecation, architecting a multi-threaded engine that partitions and mutates datasets, and orchestrates high-concurrency external service calls, to quantify data fidelity gaps.

Data Modeling 2023

- Re-architected a production DynamoDB data model using an access-pattern-driven key design to preserve historical state, eliminating update-driven data loss and strengthening auditability.
- Designed a backward-compatible migration plan using idempotent one-time backfill augmented by stream-driven change propagation for live updates, to achieve zero-downtime production migration.

MLE Execution Platform 2022

- Built the 0→1 AWS infrastructure foundation for a production optimization service, establishing a reproducible Infrastructure-as-Code (IaC) framework that standardized multi-environment release cycles and governance.
- Engineered a greenfield event-driven orchestration engine (EventBridge, Lambda, ECS) to execute containerized scientist-authored Python workflows, enabling parallel task runs that reduced end-to-end runtime by 33%.

EDUCATION

Bachelor of Computer Science, Honours, Co-op	January 2016 - June 2020
University of Waterloo, Canada	